

Imperial College London

Sep 2023 - Jun 2027

MEng Computing, currently in 4th year. Achieved Dean's List in Year 1 and 3, First Class in Year 1-3.

Studied, among other modules, Software Engineering Design, Deep Learning, Computational Techniques, Compilers, Algorithm Design and Analysis + Data Structures, Natural Language Processing, Computer Vision.

Newcastle Royal Grammar School

Sep 2016 – Jun 2023

Advanced Level (A Levels) - A* in all subjects (Maths, Further Maths, Physics and Computer Science)

EXPERIENCE

Tech Intern at Optiver

Apr 2026 - Sep 2026

- Designed and built a mass migration platform to make audited deterministic changes to >7000 repositories safely.
- Set up ingestion for multiple external datasets and improved upon the internal ingestion framework to be more scalable and maintainable, delivering a \$1 million+ upside per year, using Python and the FIX 4.4 protocol.
- Took ownership of blockers and areas of improvement outside the immediate domain of my projects, for example improving performance of the internal database schema migration tool for large migrations.

Co-Founded LineupsValorant

Jan 2021 - present

Co-founded and grew [LineupsValorant](#), a website to help Valorant players find strategies, to >70,000 monthly active users. Implemented user session management, custom strategy uploads, strategy ranking and an in-game overlay.

Software Engineering Intern at The Trade Desk

Jun 2025 - Sep 2025

Created a developer notification service, adopted company-wide, to address common complaints on internal tooling, delivering alerts for rollout changes and for actions requiring review. Built with C#, Marten DB, React.

Software Developer at Newcastle University

Dec 2021 - Sep 2024

Collected requirements and led development of software to aid studies at the university, increasing participation and engagement for 100s of patients. Primarily created with AWS EC2, BLE devices and Flask.

INVOLVEMENT

Undergraduate Teaching Assistant

Sep 2025 - Dec 2025

Supported >300 2nd year Computing students in lab sessions, helping debug issues and guide them to cleaner solutions.

DoCSoc Committee Events Officer

Aug 2026 - Present

Planned and ran events for Imperial's largest student society by sponsorship, including coordinating with venues.

PROJECTS AND ACHIEVEMENTS

LineupsValorant Computer Vision Location Classification

Jul 2026

Built and trained a lightweight CNN classifier to process frames from Valorant and detect where players are standing to show relevant lineups nearby, using PyTorch, ONNX JS runtime and LLM assisted dataset labelling.

Orchestrating Agent Harness

Jun 2026 - Sep 2026

Built an agent to oversee and interacts with multiple Claude Code sessions on my behalf and schedules recurring tasks (e.g. daily web searches). Python, FastAPI, Discord API, SQLite.

Imperial Kotlin AI Pawn Race 4th Place

Dec 2023

Developed an AI bot to participate in a modified pawn race, requiring identification and self-learning of relevant AI techniques. Used minimax algorithm with alpha-beta pruning to achieve a performant bot.

PintOS Mini Operating System

Sep 2024 - Dec 2024

Worked on a skeleton OS called PintOS, integrating MLFQ scheduling, user programs and virtual memory.

WACC Compiler

Jan 2025 - Mar 2025

Designed and built a compiler from scratch for a basic C-like programming language called WACC, including designing an object oriented extension to the language with basic inheritance with sub-classing.. Implemented using Scala and Parsley, a parser combinator library.

First and Third Year Computing Dean's List (Top 10%)

Jun 2024, Jun 2026

SKILLS

Languages	Java, C#, C, Python, Scala, Kotlin, TypeScript/JavaScript, HTML/CSS
Technologies	Git, SQL/SQLite/PostgreSQL, AWS EC2, React, Vercel, Next.js, Flask, GraphQL, PyTorch, ONNX